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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: Methods Ethical approval This study was approved by the Ethics Committee of the Chinese Peoples Liberation Army General Hospital (No. KY-2021-11-24-1).
Consent: All patients signed a written informed consent form.
Inclusion and Exclusion Criteria
not detected.
Attrition
Primary human EC specimens and cell lines A total of 51 cases of primary ESCC were surgically resected and eight cases of normal esophageal (NE) mucosa were collected from non-cancerous patients by biopsy under endoscopy.
Sex as a biological variable
not detected.
Subject Demographics
Age: Xenografted mouse model KYSE150 and KYSE30 cells stably infected with the pLenti6-PAX5 vector or empty pLenti6 vector (as control) were subcutaneously injected into the right dorsal flanks of 4-week-old male BALBc nude mice (3 x 106 cells in 0.15 mL phosphate-buffered saline) (n = 8 per group).
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
Each experiment was repeated three times.

Number: Each experiment was repeated three times.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
After 40 min of incubation with DNase at 37degC, cells were stained with a FITC-conjugated anti-BrdU monoclonal antibody (BD Bioscience Pharmingen Inc.). 7-Aminoactinomycin was added to each sample just prior to flow cytometric analysis using a FACSCalibur flow cytometer (BD Bioscience Pharmingen Inc.).	anti-BrdU		
35 The antibodies used in the experiments were rabbit anti-PAX5 (12709, 1:1000), mouse p53 (1C12) (2524, 1:1000), rabbit poly adenosine disphosphate-ribose polymerase (PARP) (46D11) (9532, 1:1000), rabbit anti-caspase-3 (9661, 1:1000), rabbit anti-caspase-7 (8438, 1:1000), rabbit anti-caspase-8 (9496, 1:1000), rabbit anti-caspase-9 (7237, 1:1000), PARP (5625, 1:1000) (Cell Signaling Technology, Danvers, MA, USA) rabbit anti-p53 antibody (ab131442, 1:1000), rabbit anti-pro Caspase-3 antibody (ab32150, 1:1000), rabbit anti-pro Caspase-7 antibody (ab32067, 1:1000), rabbit anti-Pro Caspase-8 antibody (ab108333, 1:1000), rabbit anti-pro Caspase-9 (phospho T125) (ab138412, 1:500), and rabbit anti-PAX5 antibody (ab109443, 1:1000) (Abcam, Cambridge, UK).	anti-PAX5		
	PARP		
	anti-caspase-3 (9661		
	anti-caspase-7 (8438		
	anti-caspase-8		
	anti-caspase-9		
	anti-pro Caspase-3		
	anti-pro Caspase-7		
	anti-Pro Caspase-8		
	anti-pro Caspase-9 (phospho T125		
	ab109443		
Abcam anti-PAX5 antibody (Abcam) was diluted at 1:250 and Abcam anti-p53 antibody (Abcam) was diluted at 1:100.	anti-p53		
Experimental Models: Cell Lines			
Bromodeoxyuridine (BrdU) proliferation assay and flow cytometry BrdU proliferation assays of KYSE150, KYSE30, and KYSE450 cells were carried out using a fluorescein isothiocyanate (FITC) BrdU Flow Kit (BD Bioscience Pharmingen Inc., San Diego, CA, USA).	KYSE450		Suggestion: RRID:CVCL_1353(link)
KYSE150 and KYSE30 cells were transfected with pcDNA3.1-PAX5	KYSE30		Suggestion: RRID:CVCL_1351(link)

or pcDNA3.1 (1 x 105 cellswell) in 24-well plates and cotransfected with the luciferase reporter plasmid (0.1 ngwell) and plasmid Renilla luciferase (pRL)-cytomegalovirus (CMV) (CMV promoter driving the renilla luciferase reporter) vector (2.5 ngwell) using Lipofectamine 3000.			
ChIP-PCR was performed to confirm the presence of PAX5 bound on the p53 promoter, and immunoprecipitated DNA from the KYSE150-vector, KYSE150-PAX5, KYSE30-vector, and KYSE30-PAX5 cells were used as templates.	KYSE30-PAX5		
Experimental Models: Organisms/Strains			
Xenografted mouse model KYSE150 and KYSE30 cells stably infected with the pLenti6-PAX5 vector or empty pLenti6 vector (as control) were subcutaneously injected into the right dorsal flanks of 4-week-old male BALBc nude mice (3 x 106 cells in 0.15 mL phosphate-buffered saline) (n = 8 per group).	BALBc nude		
Recombinant DNA			
Plasmid construction and transfection PAX5 expression vectors were subcloned into a pLenti6 lentiviral vector (Thermo Scientific, Wilmington, DE, USA).	pLenti6		Suggestion: RRID:Addgene_35637(link)
pCMV6 and pcDNA3.1 plasmids (Era Biotech, Shanghai, China) were used for transient transfections.	pCMV6		Suggestion: RRID:Addgene_39857(link)
	pcDNA3.1		Suggestion: RRID:Addgene_100544(link)
We inserted these molecules into the plasmid genepharma U6 (pGPU6)green fluorescent protein (GFP)Neo vector (GenePharma, Shanghai, China) that was then used to transfect KYSE450 cells.	GFP)Neo		
KYSE150 and KYSE30 cells were transfected with pcDNA3.1-PAX5 or pcDNA3.1 (1 x 105 cellswell) in 24-well plates and cotransfected with the luciferase reporter plasmid (0.1 ngwell) and plasmid Renilla luciferase (pRL)-cytomegalovirus (CMV) (CMV promoter driving the renilla luciferase reporter) vector (2.5 ngwell) using Lipofectamine 3000.	pcDNA3.1-PAX5		
	Renilla luciferase (pRL)-cytomegalovirus		
Xenografted mouse model KYSE150 and KYSE30 cells stably infected	pLenti6-PAX5		

with the pLenti6-PAX5 vector or empty pLenti6 vector (as control) were subcutaneously injected into the right dorsal flanks of 4-week-old male BALBc nude mice (3 x 106 cells in 0.15 mL phosphate-buffered saline) (n = 8 per group).			
Software and Algorithms			
Ten cytosine guanine dinucleotide (CpG) sites that spanned -297 and -136 bp regions were evaluated, which included the region analyzed by MSP. 5-Aza-2-deoxycytidine (DAC) treatment Cells were treated with 2 mmolL DAC (Sigma, St. Louis, MO, USA) in growth medium.	Sigma		Suggestion: sigma-tau (RRID:SCR_000488)(link)
SPSS 22.0 (IBM) was used for all statistical analyses.	SPSS		Suggestion: SPSS, RRID:SCR_002865(link)
For xenografted tumors, the tumor volume growth curve was drawn using Graph Pad Prism 5.0 (GraphPad Software Inc., San Diego, CA, USA).	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Relationships between clinicopathological characteristics and methylation status were analyzed using the chi-squared test and Fishers exact test.	chi-squared test
	Fishers exact test
Continuous variables with normal distribution were assessed by the independent two-sample t-test.	t-test
Oligonucleotides	
These primers were as follows: PAX5-MF: 5-AAATAAAAATTCGGTTTGCGTT C-3 (M-forward), PAX5-MR: 5-AAACATACGCTTAAAAATCGCG-3 (M-reverse) PAX5-UF: 5-TAAAAATAAAAATTTGGTTTGTGT TT-3 (U-forward), PAX5-UR: 5-TTAAAACATACACTTAAAAATCAC A-3 (U-reverse).	5-AAATAAAAATTCGGTTTGCGTT C-3
	5- AAACATACGCTTAAAAATCGCG-3
Scramble shRNAs were sense: 5-UUC UCC GAA GYC ACG UTT-3 and antisense: 5-ACG UGA CAC GUU CGG AGA ATT-3).	5-UUC UCC GAA GYC ACG UTT-3
	5-ACG UGA CAC GUU CGG AGA ATT-3

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